Jackson Baxter

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SELECTED QUANTITATIVE SYSTEMS PROJECT

TickForge — Aug 2026
Independent Quantitative Systems Project — github.com/jacksonbaxter/tickforge

- Built a deterministic C++23 market-microstructure engine that decodes Nasdaq TotalView-ITCH 5.0 data, reconstructs Level-3 order books, derives exact L1 features and forward labels, and evaluates latency- and cost-aware aggressive-IOC execution.
- Designed transactional books with expected-O(1) order-reference lookup, ordered price levels, stable FIFO queues, canonical state hashing, and checked fixed-point/rational arithmetic for market and P&L values.
- Exposed generation, replay, feature, and simulation workflows to Python through pybind11; replayed a 144,002-message, 4.8 MB synthetic corpus in 5.79–6.17 ms across seven local Apple Silicon samples.
- Established Linux/macOS CI for CTest/pytest, ASan/UBSan, and native libFuzzer smoke tests; added deterministic fault injection and locally validated the SPSC queue under TSan with one million ordered transfers.

EXPERIENCE

Curatus — Jul 2025 - Present
Software Engineer, Analytics and AI — Provo, UT

- Designed and implemented C++23 scoring and certification services for high-volume healthcare provider datasets using ODBC/SQL, multithreaded worker queues, memory-gated batch prefetch, per-thread caches, deterministic ordering, and fail-safe shutdown.
- Optimized hot paths and storage/data-processing architecture, reducing database volume 90% while preserving data fidelity; added deadlock retries, worker exception propagation, and unit, integration, end-to-end, smoke, and performance testing.
- Architected a Python/LangGraph mapping platform for millions of weekly provider records, reaching 96% data quality with schema and business-rule validation, A/B tests, regression gates, rollout controls, and rollback.

Lawrence Livermore National Laboratory — Sep 2024 - Apr 2025
Software Engineer / AI Engineer Intern — Remote

- Led a team of three engineers building a research-document RAG system, reducing average search time from about two hours to under five seconds.
- Built event-driven AWS Lambda, SQS, S3, and OpenSearch ingestion/indexing workflows, improving processing throughput 40% and retrieval accuracy 35%.

Brigham Young University IT — Sep 2021 - May 2025
Student IT Lead — Provo, UT

- Led Linux, network, database, and server upgrades that improved query performance and reliability 15%; coordinated student technicians and documented support procedures.

EDUCATION & CERTIFICATIONS

Georgia Institute of Technology — Incoming Aug 2026
Master of Science: Computer Science (OMSCS), Computing Systems — Online

Brigham Young University — Graduation - Apr 2025
Bachelor of Science: Computer Science (Emphasis in Data Science and Machine Learning) — Provo, UT

AWS Academy Graduate — Jan 2025
Cloud Foundations and Machine Learning Foundations

SKILLS

Programming Languages: C++23, Python, SQL, C#, Rust
Quantitative Development: Market data, Nasdaq ITCH 5.0, Level-3 order books, feature engineering, research simulation/backtesting, fixed-point arithmetic, data structures/algorithms
Systems Engineering: Linux, multithreading/concurrency, SPSC queues, pybind11, CMake, Git/GitHub Actions, CI/CD, ASan/UBSan/TSan, libFuzzer, performance optimization
Cloud & Data: AWS (Lambda, S3, SQS, OpenSearch, Bedrock), Docker, PostgreSQL, ODBC, Pandas, NumPy